

S - size in B -batch

S_{out} - size out

Q - 4 bytes (FP32)

k - kernel, s - stride, p - padding

C_{in} , C_{out} - Channels _{in/out}

W - layer's weights

$A_{in/out}$ - input/output tensor

CNN

Bias = False !!!

$$S_{out} = \left\lfloor \frac{S + 2P - k}{s} \right\rfloor + 1$$

$$Flops = 2BS_{out}^2 k^2 C_{in} C_{out}$$

$$W = k^2 C_{in} C_{out} Q$$

$$A_{in} = S^2 C_{in} B Q$$

$$A_{out} = S_{out}^2 C_{out} B Q$$

$$Bytes\ Moved = A_{in} + A_{out} + W$$

$$Peak = A_{in} + A_{out}$$

Max Pool

$$Flops = 0 *$$

$$W = 0$$

$$A_{in} = S$$

$$S_{out}, A_{in}, A_{out} = C_{in}$$

$$Bytes\ Moved = A_{in} + A_{out}$$

* Не проскакивал Max операций, сложна срабатывает графика. истрачивая время, но не не что мы делаем.

ReLU

inplace = True

$$\text{Flops} = 0^*$$

$$W = 0$$

0 comparisons are parallel

$$\text{Peak} = A_{in}$$

Global avg Pool

$$\text{Flops} = 0^*$$

$$W = 0$$

$$A_{in} = S^2 C B Q$$

$$A_{out} = C B Q$$

Linear ($F_{in} \rightarrow F_{out}$)

$$S_{out} = 1$$

$$\text{Flops} = 2 B F_{in} F_{out}$$

$$W = (F_{in} F_{out} + F_{out}) \cdot Q \rightarrow \text{Bias}$$

$$A_{in} = F_{in} B Q$$

$$A_{out} = F_{out} B Q$$

$$\text{Peak} = A_{in} + A_{out}$$

UToFO

F - num nodes

$$\text{Flops}(S, B) = \sum_f^F \text{Flops}_f(S, B)$$

$$\text{Memory}(S, B) = \sum_f^F w_f + \text{Max}(\text{Peak}_f(S, B) \mid F)$$

$$\text{Latency}(S, B, \theta) = \theta_{\text{startup}} + \text{Max}\left(\frac{\text{Flops}}{\theta_{\text{flops}}}, \frac{\text{Memory}}{\theta_{\text{memory}}}\right)$$

$$\text{Bytes Moved}(S, B) = \sum_f^F \text{Bytes moved}_f(S, B)$$

$$\text{Latency}(S, B, \theta) = \theta_{\text{start}} + \text{Max}\left(\frac{\text{Flops}(S, B)}{\theta_{\text{flops}}}, \frac{\text{Bytes Moved}(S, B)}{\theta_{\text{bandwidth}}}\right)$$

$$\text{Energy}(S, B, \theta) = \theta_{\text{idle}} \cdot \text{Latency}(S, B, \theta) +$$

$$\theta_{\text{per Flop}} \cdot \text{Flops}(S, B) +$$

$$\theta_{\text{byte}} \cdot \text{Memory}(S, B) +$$

$$\theta_{\text{byte}} \cdot \text{Bytes Moved}(S, B)$$